EMMANUEL SAMPSON

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EDUCATION

William & Mary, Williamsburg, VA *Expected May 2026*
BS Physics (Engineering Physics Concentration); BA Philosophy
Honors: Catron Scholar, Bosanko Scholar, Sharpe Community Scholar, Philosophy Departmental Honors

EXPERIENCE

Engineer Intern *May 2025 - Present*
WEINERT Fiber Optics Inc, Richmond, VA

- Designed AI-powered computer vision systems in Python/C++ for automated fiber-optic defect detection
- Developed CNNs using HPC clusters to identify microscopic surface defects for medical laser systems

Physics Research Assistant *October 2023 - Present*
Applied Research Center Core Labs, Newport News, VA

- Collaborated with CERN, NIST, and CALTECH on low-angle reflectivity experiments for particle detectors
- Assembled TRIFT 4 time-of-flight mass spectrometry instrument and developed custom data acquisition systems
- Operated advanced instrumentation: Hitachi S-4700 FE-SEM, Bruker ICON AFM, HIROX KH-7700, Raman Spectrometer

Student Makerspace Engineer *October 2022 - November 2023*
W&M Makerspace, Williamsburg, VA

- Managed 1,000+ service requests via Jira; trained 200+ students on laser cutters, 3D printers, CNC mills
- Pioneered 3D scanning laboratory for digital preservation of anthropological artifacts

PROJECTS

AI Fiber-Optic Defect Analyzer • *WEINERT Fiber Optics* *May 2025 - Present*
Built custom microscope with computer vision for automated defect detection, enhancing medical device transmission

Bird Sensor Technology • *W&M Hackathon 1st Place Winner* *March 2024*
Developed IoT sensor detecting bird-window collisions via frequency analysis and electromagnetic signatures

WaveFinderAI • *Hackathon 2nd Place Winner* *January 2025*
Created ML tool to detect microplastic contaminants in fluids using reflective frequency analysis

Atomistic Spin Dynamics Modeling • *Research Project* *October 2023 - Present*
Investigating Fe-Ni and Co-Fe alloys as cost-effective alternatives to rare-earth semiconductors using VAMPIRE

LEADERSHIP & ACTIVITIES

- **President:** Robotics Club (2023-25), Data Science Society (2024-25), Tech Executive Association (2024-25)
- **Creator & Host:** W&M's inaugural STEM Symposium - 15 speakers, 150+ attendees
- **Outreach Director:** Society of Physics Students, ACM
- **Awards:** W&M Hackathon Winner (2024, 2025), W&M Spirit Scholarship, National Residence Hall Honorary

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Java, JavaScript, HTML/CSS, SQL, Arduino
Engineering Software: Autodesk Fusion360, AutoCAD, SolidWorks, KiCad, Multisim, COMSOL, LabVIEW
Data Analysis: Tableau, Power BI, ArcGIS, SPSS, Jupyter, High Performance Computing (HPC)
Specialized: Computer Vision & CNNs, Fiber Optics, Signal Processing, RF Engineering, PCB Design
Instrumentation: SEM/TEM, AFM, TOF-SIMS, Raman Spectroscopy, Surface Profilers, Cleanroom Protocols